1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L35 – Data Analysis**

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4 **Empirical Study Stages**

- Broadly, four stages
  - Identification of research question(s)
  - Determination of variables
  - Design of experiment
  - Analysis of empirical data

5 **Understanding the Stages**

- We discussed first THREE stages
- In this lecture, we shall discuss last stage (data analysis)

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Data analysis

7 **Basic Idea**

- Consider study to compare text entry speed of interfaces - (for RQ3)
- Let us decide to make use of twelve participants
- Also, instead of two, let's assume we compared text entry speed of our design with another 11 interfaces (total 12 interfaces)
- We performed a repeated-measure experiment and used Latin Square method for counterbalancing
- We have 144 data items (12×12 = 144)

8 **Basic Idea**

- A portion of data shown in Table

9 **Basic Idea**

- What to do with this data?
- We can use this for drawing some conclusion on the efficacy of our design (in terms of text entry speed)
- However, the data might be misleading

10 **Basic Idea**

- We made use of twelve interfaces - represent only a fraction of all possible text input interfaces
- Twelve participants are also a very small fraction of all the potential users (even if we consider specific demographic profiles)

11 **Basic Idea**

- We are dealing with *samples* rather than the actual *population*
- Where is the guarantee the observations were not due to *chance*

12 **Basic Idea**

- If we conduct another study with a completely different set of participants and interfaces, we may end up with a completely different data set leading to a different conclusion altogether

13 **Basic Idea**

- How likely is that possibility - we need to answer this question first
- Points to "statistical significance" of the data

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Statistical significance

15 **Fundamental Concern**

- We work with samples BUT wants to draw conclusion on larger population
- Thus, we wish to conclude if the result is applicable for *any user* and *any text input interface*
  - Rather than applicable for only the twelve participants and interfaces

16 **Fundamental Concern**

- Necessary to determine *nature* of data
  - Is the data occurred *by chance*
  - Or due to *specially designed test conditions* (the interfaces) - often termed as "treatment"

17 **Basic Idea**

- Statistical significance tests answer the question
- If we perform significance test on data and find the *statistic* is *significant* with $p<0.05$ (to learn soon), we can say with *confidence* that the data is due to the "treatment" and not by chance in 95% of the times

18 **Basic Idea**

- Our starting point is a *statistic*
- In the context of user-centric research, it is the *mean* (or *average*) of a group of data items (the sample)

- Typically, we are interested to test the significance of the *difference* between the means of several groups of data (in most situations)

19 **Basic Idea**

- Ex - let us assume only two interfaces (as originally done in RQ3)
- There are twelve participants as before
- We get two "groups" of text entry speed data

20 **Basic Idea**

21 **Basic Idea**

- The *mean*, along with other measures such as *median*, *mode*, *variance* and *standard deviations*, are *descriptive statistics*
- Can reveal many things in themselves
  - E.g., the two means reveal that our design provides faster text entry than MS Word!

22 **Basic Idea**

- Can't have such general conclusion
- Same reason - where is the guarantee that the difference shall remain the way it is (i.e., group mean for our design is greater than other group mean)?

23 **Basic Idea**

- We are not concerned about absolute values (may change)
- Rather about relative difference between them
- The only way to be confidant of the reliability of the observations is to go or the significance tests

24 **Basic Idea**

- Significance tests are performed for hypothesis testing

25 **Basic Idea**

- Ex – consider the research question RQ3
  Does the new interface let me enter text "faster" than MS Word?
- Corresponding hypotheses (null and alternative)
  H_0: O*ur design is not faster than MS Word.*
  H_1: O*ur design is faster than MS Word*.

26 **Basic Idea**

- With the significance test, we try to refute the null hypothesis
- Let us try to understand the process with one simple (probably the simplest) statistical significance test

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Example – t-test

28 **Paired Sample t-test**

- If you perform the test, you are likely to get something like Table

29 **Paired Sample t-test**

- Although everything in Table is important, we do not report all these information
- Difference of the means is found to be statistically significant - we simply report this fact

30 **Paired Sample t-test**

- With a specific format
  t(11) = 2.70, p < 0.05, statistically significant

31 **Paired Sample t-test**

- In the format, we use the lowercase 't'
- After it, we put degrees of freedom (df) value (11) within parenthesis (without any space in between)
- This is followed by an '=' symbol
- Followed by the t-statistic (2.70)
- Then we put a comma and a space character
- Finally, we write the 'p' value as "p < 0.05"

32 **Paired Sample t-test**

- The 'p' value roughly indicates the probability that the data occurred by chance
- The value '0.05' is a pre-defined value
- It indicates that the probability of getting the results by chance is about 5%

33 **Paired Sample t-test**

- Since the test result indicates difference between means is statistically significant, we reject null hypothesis

34 **Paired Sample t-test**

- It may be noted that the tests run the risk of TWO types of errors

35 **Paired Sample t-test**

- Type I error (also known as the α error or "false positive")
  - Occurs when we reject a null hypothesis, which is true and should not be rejected
  - To avoid Type I errors, we typically use a very low value of p (e.g., p < 0.05)

36 **Paired Sample t-test**

- Type II error (also known as the β error or "false negative)
  - Indicates a situation where we do not reject a null hypothesis although it is false and should have been rejected
  - To avoid, it is generally recommended to go for larger sample sizes

37 **Note**

- Many important terminologies and issues (e.g., *p-value*, *two-tailed distribution*) we mentioned in the passing
  - Our aim is to introduce the idea - for more details, reference may be consulted

38 **Note**

- We also did not explain actual calculations - not necessary to perform by yourself at all
  - Instead, you can utilize any statistical package such as the SPSS™

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Techniques for Significance Test

40 **Techniques for Tests**

- t-statistic is not the only way to test for statistical significance
- There are, in fact, a large number of tests
- Very important to choose the right test method

41 **Techniques for Tests**

- Broadly TWO categories
  - Parametric
  - Non-parametric

42 **Parametric Tests**

- t-test is one of the many methods collectively known as the *parametric* methods
- Applicable subject to the fulfilment of THREE conditions

43 **Parametric Tests - Conditions**

- Data should come from a "normally distributed" population
- We should use at least an interval scale (with equally-spaced intervals) of measurement for the dependent variable (a ratio scale is even better)
- Variance in the groups of data should be approximately equal

44 **Parametric Tests**

- Did we do the right thing by applying the t-test on our data as explained earlier

45 **Parametric Tests**

- We used a ratio scale to record text entry speed (CPM)
- Thus, we are not violating the second condition for a parametric test – what about the other conditions
- Let's make an assumption

46 **Assumption**

- Text entry speeds follow a "normal" (Gaussian) distribution
- Distribution refers to entire population – we plot the speed by *all* the users (if that is possible)
  - Our sample data (speeds of twelve participants) drawn from a normally distributed population
  - Sample data in itself need not be normally distributed

47 **Parametric Tests**

- With these assumptions, we can perform *t-test* on our data
- Among the three conditions, the 2nd and 3rd are easy to verify
- It may be difficult to know the nature of the population data (the 1st condition)

48 **Parametric Tests**

- If you are not sure, you can go for additional tests such as the Shapiro-Wilk or the Kolmogorov-Smirnov test
  - Can reveal if the sample data is taken from a normally distributed population

49 **Non-Parametric Tests**

- If your data do not support any one of the above three condition, you should go for the *non-parametric* tests of significance

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52 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
  - E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

Chapter 7